

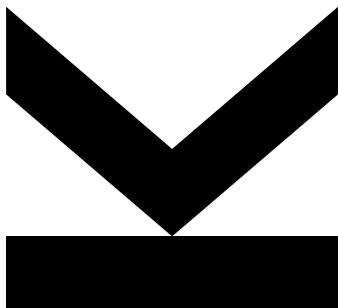
Author
Andrea Kochova

Submission
**Institute of
Biochemistry**

Thesis Supervisor
**Univ.-Prof., PhD.
Frans A. A.
Mulder**

04.2026

Comparative Analysis of Structure and Ligand Bindings in Haemoglobin Variants



Kochova Andrea
+359899226265
andreakochova964@gmail.com
k12232309

SWORN DECLARATION

I hereby declare under oath that the submitted Bachelor Thesis has been written solely by me without any third-party assistance, information other than provided sources or aids have not been used and those used have been fully documented. Sources for literal, paraphrased and cited quotes have been accurately credited.

The submitted document here present is identical to the electronically submitted text document.

Linz, 04.2026

A handwritten signature in black ink, appearing to be 'H. K.', is written on a light gray rectangular background.

Table of Contents

1. Introduction	3
1.1. Haemoglobin as a model system.....	3
1.2. Structural organisation of the haemoglobin fold.....	3
1.3. Ligand binding and allosteric regulation.....	4
1.4. Structural variation in Haemoglobin: mutations and evolutionary diversity	4
1.5. A bioinformatic approach to studying protein structure and ligand interaction.....	5
2. Work Aims	6
3. Materials and Methods	6
3.1. Dataset Curation.....	6
3.1.1. Haemoglobin proteins	6
3.1.2. Ligands	7
3.2. Computational tools	8
3.3. Molecular docking.....	8
3.3.1. Protein preparation	8
3.3.2. Ligand preparation	8
3.3.3. Blind docking simulations.....	8
3.3.4. Selection of a representative (“best”) pose	9
4. Aim-specific computational analyses.....	9
4.1. Heatmap analysis of docking behaviour	9
4.2. Ligand centroid dispersion analysis	10
4.3. Structural comparison of haemoglobin variants.....	11
5. Results	12
5.1. Dataset and docking outputs overview.....	12
5.2. Heatmap analysis of docking behaviour	12
5.2.1. Binding energy distribution.....	12
5.2.2. Cluster reproducibility throughout docking simulations.....	13
5.2.3. Combined overview of mean binding energy and cluster reproducibility patterns	14
5.3. Ligand centroid dispersion analysis	15
5.3.1. Centroid dispersion of IHP.....	15
5.3.2. Centroid dispersion of 2,3-BPG	15
5.3.3. Visualisation of centroid distributions.....	16
5.4. Structural comparison of haemoglobin variants.....	17
5.4.1. Global and heme-pocket structural deviations	17

5.4.2.	Residue-level structural variability within the heme pocket	17
5.4.3.	Spatial distribution of residue-level deviations	18
5.4.4.	Most variable pocket residues	18
6.	Discussion	19
6.1.	Conceptual framework	19
6.2.	Key findings	19
6.3.	Ligand-dependent docking behaviour	19
6.4.	Spatial constraints of ligand binding	20
6.5.	Structural basis of haemoglobin function.....	21
6.6.	Structure-function relationship	22
6.7.	Biological implications and comparison with literature.....	22
6.8.	Limitations and future directions	23
7.	Conclusion.....	23
8.	References	24

1. Introduction

1.1. Haemoglobin as a model system

Haemoglobin (Hb) is one of the most broadly studied proteins in the fields of biology, chemistry and biochemistry. It has a primary role in respiration, as it is the main oxygen-transport protein in vertebrate blood. It enables the reversible binding and release of molecular oxygen between the lungs and peripheral tissues (Olson & Phillips, 1997; Perutz, 1970).

The haemoglobin structure serves as an archetype for exploring the relationship between protein structure, ligand binding, and allosteric regulation. It is made up of four globin subunits arranged in a tetrameric $\alpha_2\beta_2$ configuration, and each globin has a heme prosthetic group. The group is comprised of a protoporphyrin IX ring coordinated to a central iron atom, which provides the chemical centre for oxygen binding – the key to allowing haemoglobin to reversibly coordinate molecular oxygen through an $\text{Fe}^{2+}\text{-O}_2$ interaction (Dickerson & Geis, 1993; Perutz et al., 1960). Furthermore, the position of the heme group within a hydrophobic pocket in the globin fold ensures correct coordination geometry and protects the iron centre from irreversible oxidation.

The structural organisation of haemoglobin also enables cooperative oxygen binding - a property whereby oxygen binding at one heme site increases the affinity of the remaining sites. This behaviour is due to conformational transitions between two principal quaternary states: the low-affinity tense (T) state and the high-affinity relaxed (R) state (Eaton et al., 1999; Perutz, 1970). It demonstrates how protein architecture can regulate ligand binding through allosteric mechanisms, since these transitions require coordinated structural rearrangements across subunit interfaces. Because haemoglobin combines well-defined structural features with measurable outputs, it serves as a model system to investigate how proteins maintain functional stability while accommodating structural variation.

1.2. Structural organisation of the haemoglobin fold

The globin fold is a highly conserved α - helical structure that surrounds , arranged around the heme prosthetic group. This is adopted by each of the haemoglobin subunits (Kendrew et al., 1960; Perutz et al., 1960). Within this fold, where the heme molecule lies, the residues from multiple helices form a well-defined binding pocket. In particular, two histidines are the most central in heme coordination: the proximal histidine, which directly coordinates the iron atom, and the distal histidine, which stabilises bound oxygen through hydrogen bonding and steric control of ligand orientation (Olson & Phillips, 1997).

The geometry of the heme pocket is crucial for haemoglobin function. Proper positioning of the iron atom with respect to the porphyrin ring and surrounding residues determines the ligand accessibility, binding affinity, and ability of the protein to undergo cooperative structural transitions. Consequently, this is the reason why the residues that form the heme-binding environment tend to be highly conserved between species and various haemoglobin variants (Hardison, 1996).

Conversely, other regions of the protein, mainly surface loops and peripheral helices, may experience more structural variation. Rather than directly coordinating with the heme group, these regions are involved in inter-subunit contacts or interactions with regulatory molecules. Thus, haemoglobin provides a good example of how proteins can contain both rigid functional cores and structurally flexible regions. Understanding the dynamics of these regions is crucial to properly interpreting the functional differences among haemoglobin variants, even in the context of different evolutionary scales.

1.3. Ligand binding and allosteric regulation

Haemoglobin activity is largely modulated by several small regulatory ligands that bind outside of the heme pocket. They indirectly regulate oxygen affinity, as they alter the equilibrium between the T and R states. One of the most important physiological regulators is 2,3-bisphosphoglycerate (2,3-BPG). It is an anionic metabolite that binds within the central cavity of the haemoglobin tetramer (Benesch & Benesch, 1967). That binding of the 2,3-BPG ligand stabilises the T-state of the protein by forming electrostatic interactions with positively charged residues located at the α - β subunit interfaces, thereby reducing oxygen affinity and facilitating oxygen release to peripheral tissues (Arnone, 1972; Benesch & Benesch, 1967; Perutz, 1983).

Other highly charged polyanions can bind in the same central cavity and act as allosteric effectors. Such an effector is inositol hexaphosphate (IHP). It binds in a similar region of the central cavity, and, as it possesses a higher negative charge density than 2,3-BPG, can strongly stabilise the T-state through electrostatic interactions (Arnone & Perutz, 1974; Shulman et al., 1975). Both of these ligands interact with structural components that arise from the quaternary organisation, rather than from the individual subunits. Therefore, changes in subunit packing or cavity geometry can influence ligand binding behaviour.

1.4. Structural variation in Haemoglobin: mutations and evolutionary diversity

Despite the highly conserved nature of haemoglobin, some occurring variants have been identified. The majority arise from single amino acid substitutions in the globin chains,

consequently influencing oxygen affinity, protein stability, or allosteric regulation (Bunn & Forget, 1986). Several characterised variants show how local structural perturbations can affect haemoglobin behaviour. An example is the substitutions in the β -globin chain that may alter inter-subunit interactions, destabilise conformational states, or modify binding sites for regulatory molecules. Furthermore, mutations near the central cavity can affect interaction with 2,3-BPG, and substitutions near the heme pocket may influence ligand binding (Thom et al., 2013).

On the other hand, ancestral proteins, reconstructed through phylogenetic methods, provide a framework for studying protein evolution. Through either experimental or bioinformatic analyses of their structures, it becomes possible to explore structural and functional properties over different evolutionary scales (Hochberg & Thornton, 2017; Thornton, 2004). Thus, comparing modern, mutant and ancestral haemoglobin variants enables the exploration of how structural variation emerges under various biochemical and evolutionary constraints.

1.5. A bioinformatic approach to studying protein structure and ligand interaction

Detailed investigation of protein-ligand interactions and conformational changes has been made possible with the advances in structural biology and computational chemistry. One of the most common bioinformatic approaches is molecular docking. It is a technique that predicts how small molecules interact with protein structures. It does that by finding all possible binding orientations and evaluating their energetic favourability (Meng et al., 2011; Morris et al., 1998).

Most docking algorithms sample ligand conformational space within a defined region of a protein structure, and each pose is scored with scoring functions, which are meant to approximate the binding free energies. On the contrary, blind docking methods further this strategy by allowing the ligands to explore the entire protein surface rather than restricting the search to a pre-defined binding site. This approach is most useful when the exact binding site is uncertain. Or when comparative analysis across multiple proteins is required, as applying the same docking parameters across multiple structures allows the observed differences in docking behaviour to be interpreted as consequences of structural variation between proteins. Additionally, it can help reveal new potential binding sites and energetic minima across the protein surface (Meng et al., 2011).

After docking, the output can be analysed using several metrics that describe the energetic as well as the spatial characteristics of the predicted binding modes. This is of great importance, as predicted binding energies provide an estimate of energetic favourability for a given pose,

while clustering of docking solutions reveals dominant binding geometries that occur across distinct docking runs. Moreover, the spatial dispersion of the docking poses provides insight into how strongly a binding site constrains the ligand positioning. Overall, using these metrics, the docking results can be interpreted not only in terms of binding affinity but also in terms of the geometric and structural constraints imposed by the protein environment (Meng et al., 2011).

2. Work Aims

The study aims to translate the conceptual understanding of haemoglobin into a quantitative, comparative analysis of protein-ligand interactions. The interplay between structural conservation, allosteric regulation, and ligand binding is explored across modern, mutant, and ancestral haemoglobin variants using a purely computational approach. Therefore, three analytical levels were defined. First, structural comparison is used to establish the degree and localisation of structural conservation within the haemoglobin fold, particularly the heme-binding pocket. Second, a centroid-based spatial analysis quantifies the extent to which ligand positioning is geometrically constrained within each protein environment. Third, molecular docking provides a systematic evaluation of ligand-binding behaviour, capturing both energetic favourability and binding mode reproducibility. Together, these aims answer the central question of this study: how structural variation within a highly conserved protein scaffold influences ligand-binding energetics and spatial behaviour.

3. Materials and Methods

3.1. Dataset Curation

3.1.1. Haemoglobin proteins

The analyses in this study were performed on a dataset comprising haemoglobin structures, representing three categories: modern proteins, mutant variants, and reconstructed ancestral haemoglobins. The dataset was designed to allow the systematic comparison of structural conservation and ligand-binding behaviour across mutational and evolutionary contexts.

We selected three experimentally determined haemoglobin structures to represent the modern haemoglobins. The structure with Protein Data Bank (PDB) identifier “1A3N” was used as the primary reference structure throughout the study due to its well-resolved geometry and extensive structural characterisation. Additionally, two other modern haemoglobin structures, with the identifiers “2DN3” and “6BB5”, were included. This approach represented alternative

experimentally observed conformational states and captured the range of structural variability among modern haemoglobin proteins.

Afterwards, four mutant variants were included in the dataset, under the identifiers: D99N, K82N, N102T, and Y35F. These variants represent single amino acid substitutions, shown with the first and last letter in the identifier, at a specific location in the sequence, represented by the number. The mutants are based on the variants in Table 1 in the journal article by Thom et al. (2013). The human adult haemoglobin sequence was retrieved from NCBI (*National Center for Biotechnology Information (NCBI)[Internet].*, 1988), and, according to the article, the single amino acid substitutions were marked. The structural models for the mutant proteins were then generated with the AlphaFold 4 web client (Abramson et al., 2024).

To investigate the structural variation across evolutionary timescales, three ancestral haemoglobin reconstructions were included in the dataset: AncA, AncB and AncMH_nat. Their sequences were obtained from previously published ancestral reconstructions of globin evolution (Pillai et al., 2020). They correspond to the ancestral α - and β -globin proteins (here AncA and AncB) and the ancestral native myoglobin/haemoglobin globin (here AncMH_nat). The reconstructed sequences are available in the NCBI database under accessions MT079115, MT079113, and MT079114.

3.1.2. Ligands

For docking analysis, a set of ligands was prepared. It included the haemoglobin prosthetic group and its variants, and physiologically relevant allosteric effectors. The heme ligand set is made up of multiple variants, representing different coordination states. The variants included heme with iron (heme_Fe), heme with oxygen (heme_O2), heme with carbon monoxide (heme_Co), and metal-free heme. They were included to evaluate how differences in coordination chemistry influence binding behaviour across haemoglobin variants. Additionally, two allosteric regulators were included, namely 2,3-bisphosphoglycerate (2,3-BPG) and inositol hexaphosphate (IHP). As mentioned above, these molecules are highly negatively charged and modulate the oxygen-binding properties. The ligand structures were obtained from the PubChem database (Kim et al., 2025) as 3D conformers or were extracted from experimentally resolved haemoglobin complexes. Before docking, all ligand structures were standardised, as described later on.

3.2. Computational tools

All computational analyses were performed using a blend of molecular modelling, docking, and data analysis tools. UCSF ChimeraX (Pettersen et al., 2020) was used to for protein structure preparation and visualisation. Ligand geometry optimisation was done with Avogadro (Hanwell et al., 2012). Molecular docking simulations were carried out with AutoDock 4 from the AutoDockTools (MGLTools) package (Sanner, 1999). Data processing and analysis of docking outputs were implemented with Python scripts and R/RStudio (Posit team (2025), n.d.).

3.3. Molecular docking

3.3.1. Protein preparation

To ensure proper docking and compatibility with the AutoDock4 engine, all protein structures were processed. First, solvent molecules were removed. Subsequently, any remaining irrelevant heteroatoms were removed to prevent interference with ligand-binding simulations. All hydrogen atoms were added, and all non-polar ones were merged. Partial atomic charges were assigned, using the Kollmann charge model. Additionally, AutoDock-specific atom types (AD4 type) were defined for all atoms. Lastly, the processed protein structures were converted and exported into PDBQT format, which encodes both atomic coordinates and docking-specific parameters required for the simulation.

3.3.2. Ligand preparation

Firstly, hydrogen atoms were added to each ligand structure, followed by geometry optimisation with Avogadro's built-in energy minimisation algorithm. Optimisation was repeated until convergence was achieved, and no further significant atomic movements were observed. To preserve charge information and atomic connectivity, the optimised ligand structures were exported in MOL2 format. Then, in AutoDockTools, they were further processed. The torsion trees were detected to identify rotatable bonds. The assignment of rotatable bonds allowed full ligand flexibility during docking simulations. Lastly, all final optimised ligand structures were exported in PDBQT format.

3.3.3. Blind docking simulations

The molecular docking simulations were performed using AutoDock 4. A blind docking approach was employed so that the search space encompassed the entire protein structure. For that reason, docking grids were generated such that the grid box fully enclosed the protein. The grid dimensions depended on the size of the respective protein structure.

After all the grid maps were generated, a Lamarckian Genetic Algorithm was chosen as the search strategy. For each protein-ligand combination, 100 independent genetic algorithm (GA) runs were performed. Default AutoDock parameters were used, with extended evaluation settings – the maximum number of evaluations was set to “long”. This ensured sufficient conformational sampling. All the information from a docking simulation was stored in the output docking log files (.dlg files).

3.3.4. Selection of a representative (“best”) pose

To sort through the predicted ligand-binding poses for each protein-ligand docking simulation, Python scripts were utilised. They were written to extract cluster-level information and define representative binding modes for each protein-ligand pair.

First, for each docking simulation, the script extracted cluster statistics, which include cluster rank, mean binding energy, lowest binding energy, and cluster population. Then, for the most precise determination of a binding mode, a rule-based selection procedure was applied. Clusters consisting of a single pose were excluded if multi-member clusters were present. In the remaining clusters, the one with the lowest mean binding energy was identified. Clusters within 0.5 kcal mol⁻¹ of this minimum were considered energetically equivalent. Subsequently, from that subset, the cluster with the highest population was selected as the representative binding pose. Lastly, the script generated a summary dataset, containing all protein-ligand combinations and their associated cluster statistics. The dataset was used as input for downstream analyses of docking behaviour and ligand spatial distribution.

4. Aim-specific computational analyses

4.1. Heatmap analysis of docking behaviour

The employment of a heatmap-based approach allowed us to obtain a global and comparative overview of ligand-binding behaviour across all haemoglobin variants. The analysis simultaneously evaluated two key aspects of docking behaviour: the predicted energetic favourability of ligand binding, and the reproducibility of binding modes.

The main input for this pipeline was the summary dataset, generated during docking result processing (2.3.4. Selection of a representative (“best”) pose). For each system, cluster-level statistics were extracted, including mean binding energy and the number of docking poses assigned to the selected cluster. Two heatmaps were constructed with Python-based data tools to visualise the extracted metrics across the full dataset. In both plots, rows corresponded to the haemoglobin variants and columns to the ligands.

The mean binding energy of the selected docking cluster for each protein-ligand pair ($n = 60$ total distinct pairs) was used to construct the first heatmap. And, as calculated by the AutoDock scoring function, the binding energies were kept in kcal mol^{-1} . The second heatmap was intended to represent cluster reproducibility. To construct it, we defined the cluster reproducibility as the fraction of docking runs assigned to the selected cluster. Since each docking simulation consisted of 100 independent GA runs, the metric was calculated as follows:

$$\text{cluster fraction} = \frac{\text{cluster population}}{100}$$

Equation 1: Calculation of cluster reproducibility

4.2. Ligand centroid dispersion analysis

To quantify the spatial variability of ligand binding, we focused on a centroid-based approach. It analyses docking poses, so the degree to which ligand binding positions are spatially constrained or dispersed within each protein-ligand system could be assessed.

For a broader scope, all predicted ligand poses generated by AutoDock ($n = 100$ per distinct pair) for each docking simulation were included. Then, for every individual docking pose, the three-dimensional coordinates of all ligand atoms were extracted. The spatial position of each ligand was then represented by its centroid, defined as the arithmetic mean of the atomic coordinates (Equation 2). In the equation, we designated the variables as follows: M is the number of atoms in the ligand, and (x_j, y_j, z_j) are the Cartesian coordinates of the atom j .

$$c = \left(\frac{1}{M} \sum_{j=1}^M x_j, \frac{1}{M} \sum_{j=1}^M y_j, \frac{1}{M} \sum_{j=1}^M z_j \right)$$

Equation 2: Centroid computation

After the computation, the outputs were parsed, and a dataset containing the centroid coordinates and associated docking energies was generated. Furthermore, to quantify the spatial dispersion, the distribution of centroid positions was investigated for each protein-ligand pair. Consequently, a summary dataset was generated, with one row per system, including the number of poses ($n = 100$), the mean radial distance ($mean(r)$), standard deviation ($sd(r)$), median radial distance, and interquartile range ($IQR(r)$). The mean radial distance was used as the primary metric, as it reflects the average deviation of docking poses from a central reference position. Additionally, the interquartile range was reported, as it is a robust measure of dispersion that is less sensitive to outliers.

4.3. Structural comparison of haemoglobin variants

To assess structural conservation and variability across haemoglobin variants, a comparative structural analysis was conducted. The use of coordinate-based alignment and root-mean-square deviation (RMSD) enabled us to distinguish globally conserved structural features from regions exhibiting increased variability.

All structural comparisons were done relative to the reference haemoglobin structure 1A3N, which acted as the structural baseline. Before alignment, protein structures were to ensure consistency throughout the dataset. Each haemoglobin structure was reduced to its β -globin subunit. Even though haemoglobin exists as a tetrameric $\alpha_2\beta_2$ complex, the mutations investigated here are located exclusively within the β subunit. Additionally, mutant models were available as single-chain structures. Therefore, restricting the analysis to the chain ensured structural comparability across all proteins while avoiding ambiguities arising from differences in quaternary structure. It should be noted that ancestral haemoglobin variants were not included due to limited representation ($n = 1$), as only AncB was deemed suitable for this analysis. Including such a limited sample would not have allowed for a meaningful assessment of structural variability at the group level.

During the standardisation process, all water molecules (solvent) and non-standard residues were removed. An exception makes the reference structure, for which the heme group (HEM) was retained. This was done so that the ligand-binding pocket could be defined. All structures were subsequently aligned to the reference using backbone $C\alpha$ atoms. Alignment was performed with the *matchmaker* algorithm of ChimeraX, as it combines sequence alignment with structural superposition.

Besides global RMSD, a local structural analysis of the heme-binding pocket was conducted. The pocket was defined in the reference structure based on residues located within two distance shells around the heme group, namely: residues within 5 Å (first shell), representing direct contacts, and residues within 5-7 Å (second shell), representing the surrounding structural environment. To ensure correct correspondence despite potential insertions or deletions, equivalent residues were identified through sequence-based residue mapping. Then, RMSD calculations were performed for this specific subset of pocket residues. This allowed focused evaluation of structural conservation within the functionally critical ligand-binding region. Lastly, to further characterise structural variability with more precision, the per-residue positional deviations were calculated for all pocket residues. For every protein structure, the

displacement of each C α atom relative to the corresponding atom in the reference was computed as the Euclidean distance between aligned coordinates. All RMSD calculations and residue mapping analysis were performed with the MDAnalysis Python library (Gowers et al., 2016). All final metrics – global RMSD, pocket-specific RMSD, and per-residue deviations- were visualised with the Python Matplotlib library (Hunter, 2007).

5. Results

5.1. Dataset and docking outputs overview

The analyses throughout the study were performed on a dataset comprising haemoglobin variants, which were separated into three categories: modern, mutant, and ancestral. The total number of haemoglobin structures was ten, distributed as follows: experimentally determined modern structures ($n = 3$), mutant variants arising from single amino acid substitutions ($n = 4$), and reconstructed ancestral proteins ($n = 3$). Additionally, a set of sex ligands was used for the docking simulations. It included four heme variants (heme_Fe, heme_O₂, heme_CO, and metal-free heme) and two allosteric effectors (2,3-BPG and IHP). The combination of ten protein structures and six ligands resulted in a total of 60 distinct protein-ligand pairs. For each system, molecular docking simulations were carried out with AutoDock 4. The 100 independent runs per system yielded a total of 6000 predicted ligand-binding poses across the entire dataset. From these simulations, representative binding modes for each protein-ligand pair were generated, as well as other cluster-based summaries.

5.2. Heatmap analysis of docking behaviour

5.2.1. Binding energy distribution

For each of the 60 protein-ligand systems, the mean binding energy of the selected docking cluster was determined and visualised in Figure 1. Binding energies are reported in kcal mol⁻¹, as computed by the AutoDock scoring function.

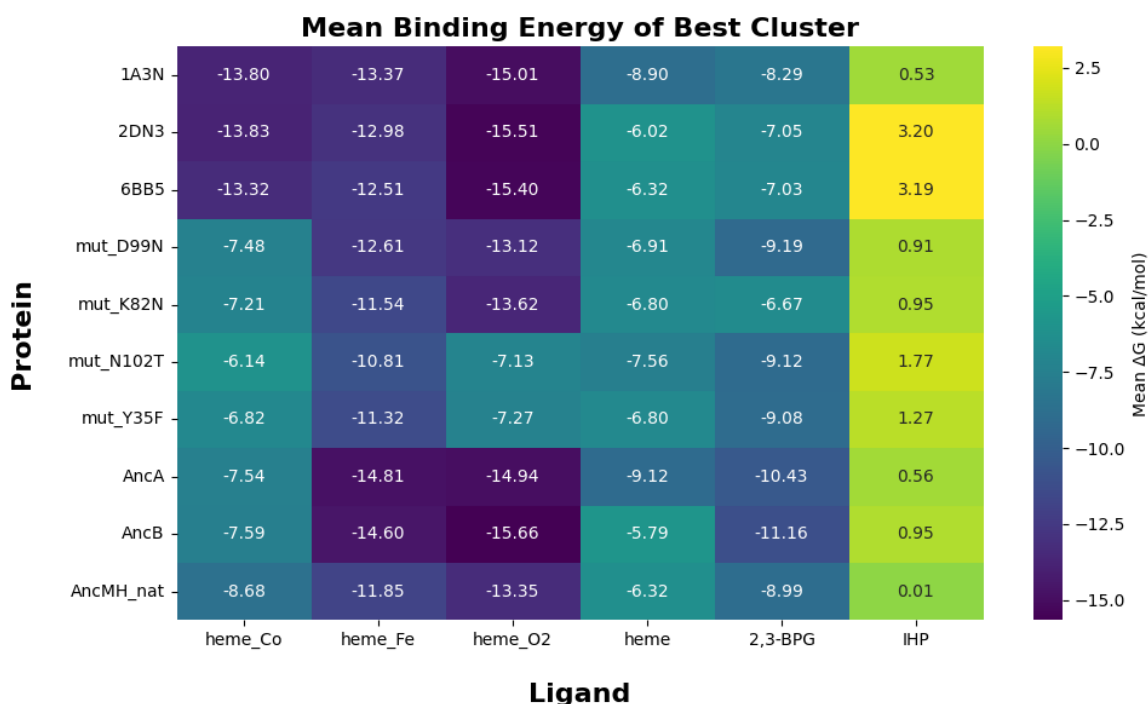


Figure 1: Heatmap of Mean Binding Energy of chosen cluster for each distinct protein-ligand system

Across all systems, the mean binding energy ranged from -15.66 to 3.20 kcal mol⁻¹. The lowest values were observed for the heme ligands, particularly heme_O2 and heme_Fe. They consistently exhibited energies between approximately -16 and -12 kcal mol⁻¹ across multiple protein structures. The most negative value of -15.66 kcal mol⁻¹ was observed for the AncB-heme_O2 system. Metal-free heme displayed less favourable energies across all protein classes, with values in the range of -9.12 to -6.02 kcal mol⁻¹.

The allosteric ligand 2,3-BPG had mean binding energies ranging from -11.16 to 6.67 kcal mol⁻¹. The more negative part of this range was largely represented by the ancestral proteins, while modern proteins had values closer to ~7 to ~8 kcal mol⁻¹. The mutant variants showed partially heterogeneous values, with relatively favourable ones of approximately 9 kcal mol⁻¹. An exception was the mutant mut_K82N, where the mean binding energy is substantially less favourable (-6.67 kcal mol⁻¹). On the other hand, the ligand IHP exhibited the widest range of values spanning from 0.01 to 3.20 kcal mol⁻¹. Across all protein classes, IHP values were consistently in the positive range. However, no coherent, uniform trends could be observed for either protein category.

5.2.2. Cluster reproducibility throughout docking simulations

The cluster reproducibility values ranged from 0.02 to 0.84, corresponding to 2-84% of docking runs converging to the same binding mode. The highest reproducibility value, 0.84, was observed for the AncMH_nat-2,3-BPG system.

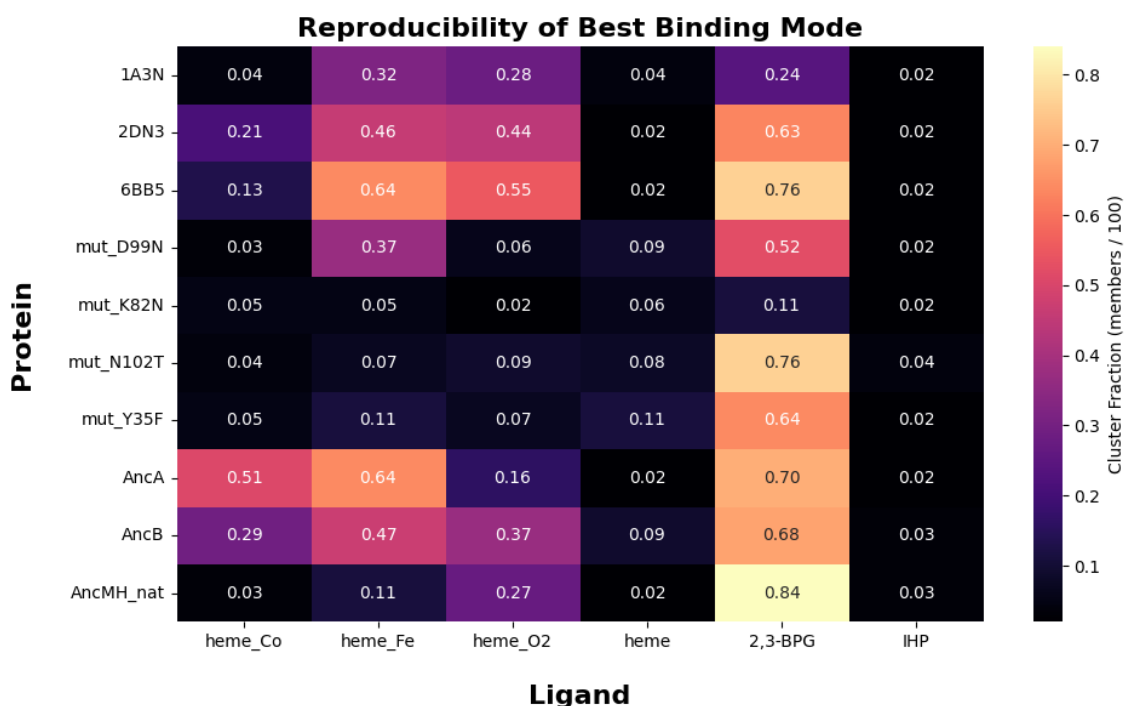


Figure 2: Heatmap of cluster reproducibility for each selected representative pose per protein-ligand system

Reproducibility varied across ligand types. The ligand 2,3-BPG consistently exhibited higher values across most proteins, typically in the range of approximately 0.52 to 0.84. For the modern protein category, 2,3-BPG values spanned approximately 0.24 to 0.76, while for the ancestral ones, values were between 0.68 and 0.84. Mutants had the broadest range, from 0.11 to 0.76.

The heme-derived ligands also had variations. For heme_Fe and heme_O2, reproducibility values for modern proteins were generally moderate, approximately 0.28 to 0.64, whereas mutant proteins had values concentrated in the lower ranges. An exception was the mutant D99N-heme_Fe system, which had a lower moderate value of 0.37. Similar to modern proteins, ancestral ones had a moderate value range from 0.11 to 0.64. The ligand heme_Co has a similar pattern with slightly lower values. Its reproducibility values were moderate in modern proteins, variable in ancestral proteins, and low in mutants. Conversely, metal-free heme exhibited consistently low reproducibility across all protein classes, with the typical range being from 0.02 to 0.11. Furthermore, IHP also showed uniformly low values across all systems (0.02-0.04).

5.2.3. Combined overview of mean binding energy and cluster reproducibility patterns

The distributions of mean binding energy (Figure 1) and cluster reproducibility (Figure 2) were compared across all protein-ligand systems. Yet, no consistent relationship between these two metrics was observed. Systems with strongly negative binding energies, especially for heme ligands, did not continuously exhibit high reproducibility. In contrast, higher reproducibility values were frequently observed for 2,3-BPG across multiple protein structures, despite the corresponding binding energies being moderate. Across the dataset, similar energy ranges were associated with variable reproducibility.

5.3. Ligand centroid dispersion analysis

5.3.1. Centroid dispersion of IHP

Centroid dispersion values for IHP were confined to a narrow range of approximately 0.015 to 0.020 Å across all haemoglobin variants, as seen in Figure 3. Modern, mutant and ancestral proteins exhibited highly similar values, with only minor deviations between individual systems, such as mutant K82N with values ~ 0.015 Å and AncMH_nat with ~ 0.020 Å. In general, no significant separation between protein classes was observed.

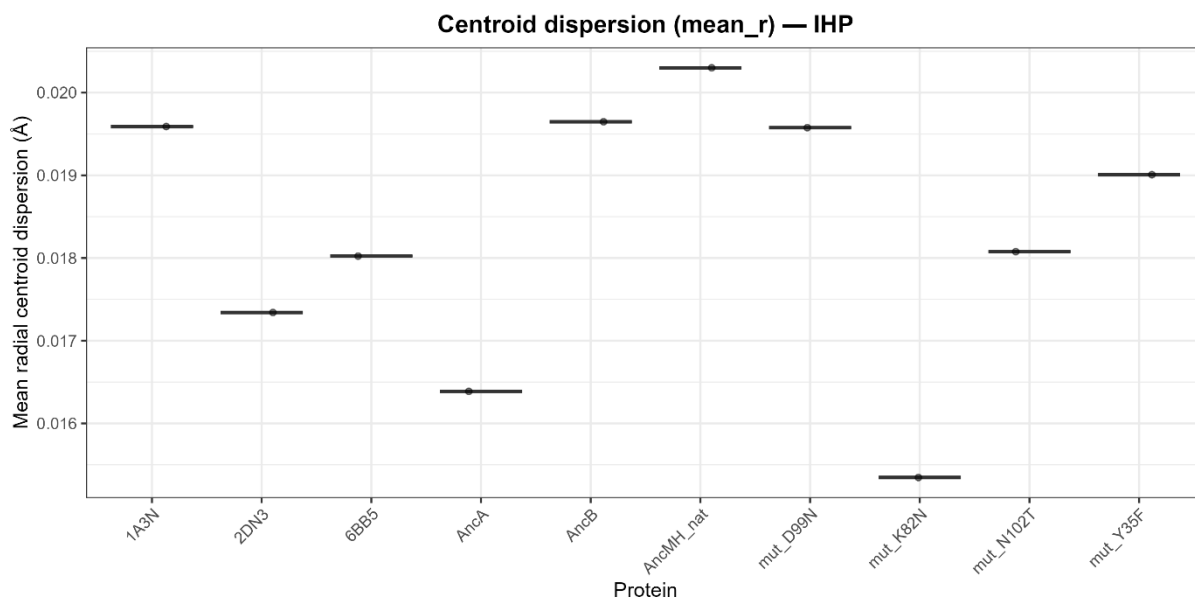


Figure 3: Mean radial distances for IHP across all protein classes

5.3.2. Centroid dispersion of 2,3-BPG

Conversely, 2,3-BPG exhibited a broader range of centroid dispersion values, ranging from approximately 0.022 to 0.057 Å (Figure 4). This corresponds to a two- to three-fold larger range compared to IHP. Modern haemoglobins had lower dispersion values (~ 0.022 to ~ 0.038 Å). Mutant and ancestral proteins exhibited higher values, such as ~ 0.052 Å for mutant K82N and ~ 0.057 Å for AncB, as well as some intermediate cases (~ 0.032 - 0.033 Å in AncA and AncMH_nat).

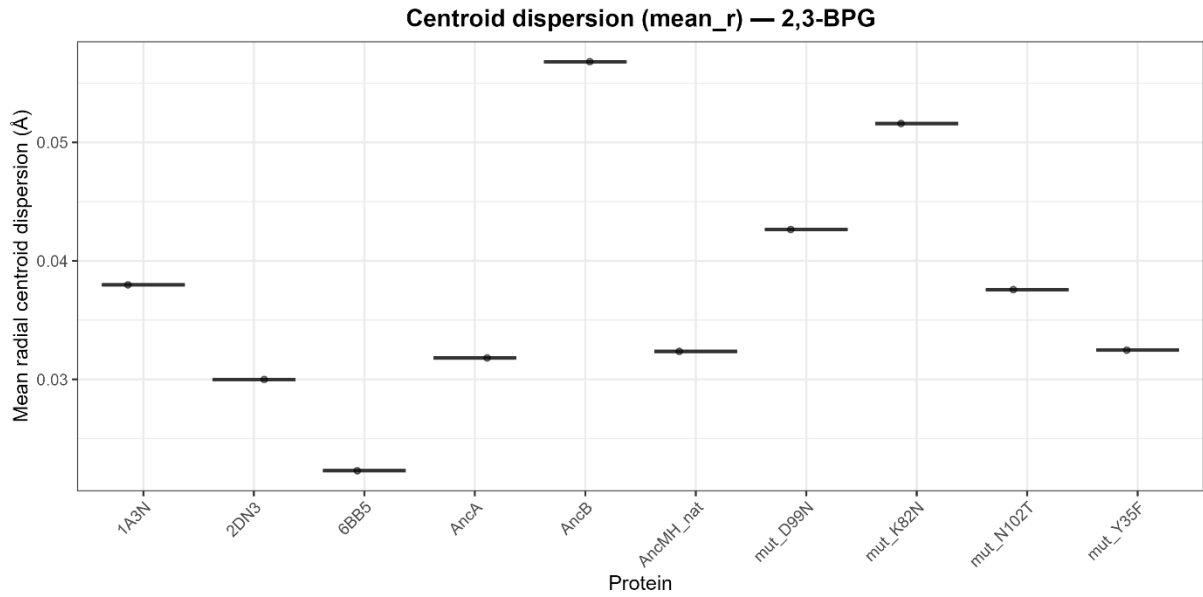


Figure 4: Mean radial distances for 2,3_BPG across all protein classes

5.3.3. Visualisation of centroid distributions

Three-dimensional (3D) visualisations of centroid positions for 2,3-BPG centroids were generated for representative haemoglobin variants: the modern structure 1A3N, the ancestral variant AncB, and the mutant variant mut_K82N (Figure 5).

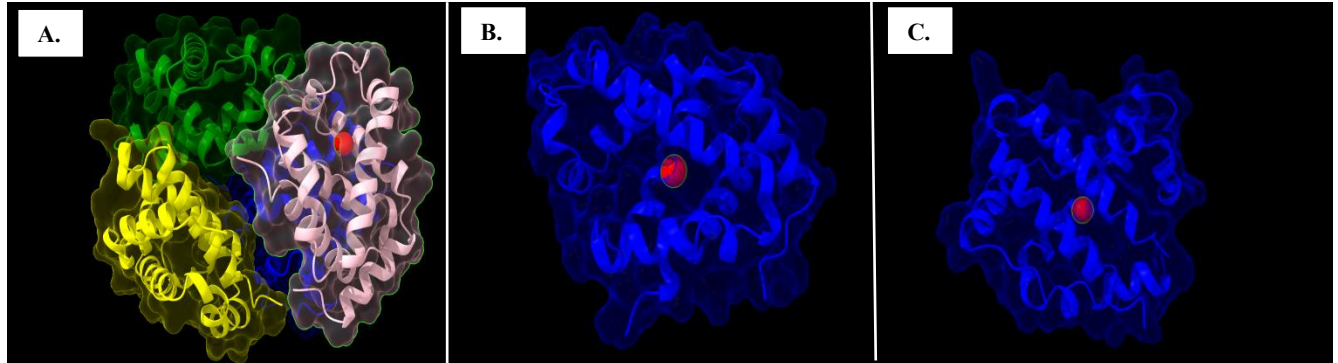


Figure 5: 3D Visualisations of centroid positions for 2,3-BPG with proteins: A. 1A3N, B. AncB, and C. mutant K82N

Across all systems, centroid positions were consistently localised within the central cavity of the haemoglobin structure. The overall spatial location of the centroid clouds remained similar, with no displacement to alternative binding regions observed. The differences between proteins were reflected in the extent of spatial spread, rather than in the position of the centroid clusters. This is seen in 1A3N, which exhibited a relatively compact centroid distribution, while AncB and mut_K82N showed broader spatial envelopes within the same cavity region.

The observed differences in centroid distribution were moderate in magnitude, as the overall range of centroid dispersion corresponds to approximately a two-fold difference in mean radial distance across systems. Therefore, the centroid clouds remained centred in the same binding region and did not form clearly separated clusters.

5.4. Structural comparison of haemoglobin variants

5.4.1. Global and heme-pocket structural deviations

Global and local structural deviations were quantified using backbone C α RMSD relative to the reference structure 1A3N (Figure 6). Global RMSD values ranged from 0.63 to 0.91 Å. The modern proteins exhibited the highest values, with 0.91 Å for 2DN3 and 0.89 Å for 6BB5. Conversely, mutant and ancestral haemoglobins had lower RMSD values, ranging from 0.63 to 0.71 Å. On the other hand, pocket RMSD values were consistently lower than global ones, spanning from 0.49 to 0.78 Å. Similarly to the pattern observed with global RMSD, here modern proteins also had the highest values of 0.78 Å (2DN3) and 0.72 Å (6BB5). In general, the relationship between global and pocket RMSD was observed to be consistent across both classes, with the pocket RMSD values remaining lower than the global ones in all cases.

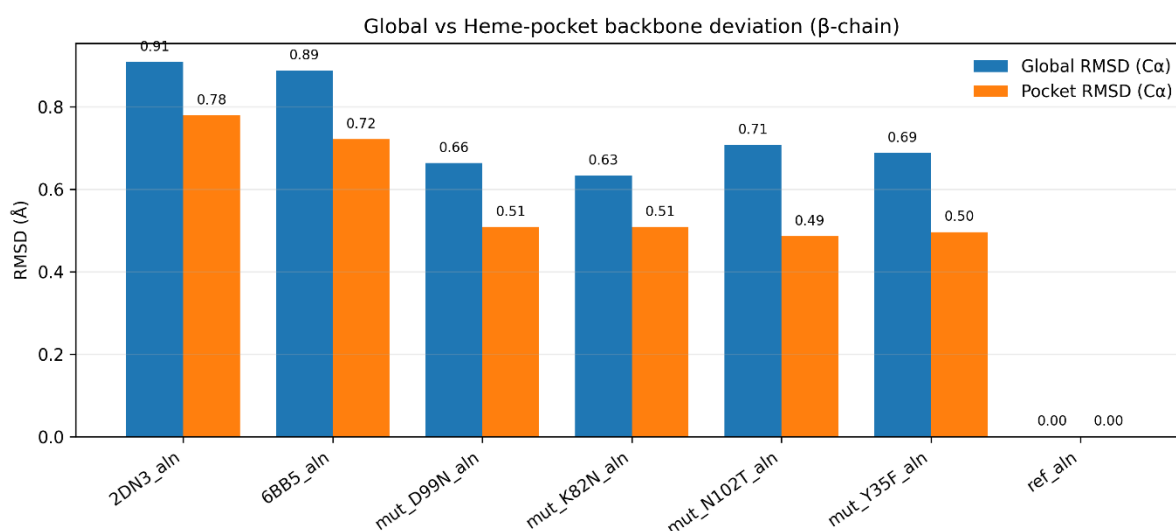


Figure 6: Global and local (heme-pocket) backbone deviation in the beta-chain, relative to the reference 1A3N

5.4.2. Residue-level structural variability within the heme pocket

Per-residue backbone deviations ($\Delta C\alpha$) were computed for all pocket residues (Figure 7). Residue-level deviations across chosen haemoglobin variants were within the range of ~0.2 to 1.7 Å. Modern haemoglobins showed moderate deviations across multiple residues, while the mutant and ancestral ones had similar patterns of slightly lower overall values. Overall, the majority of pocket residues displayed low to moderate deviations, with values for most structures falling below ~1.0 Å.

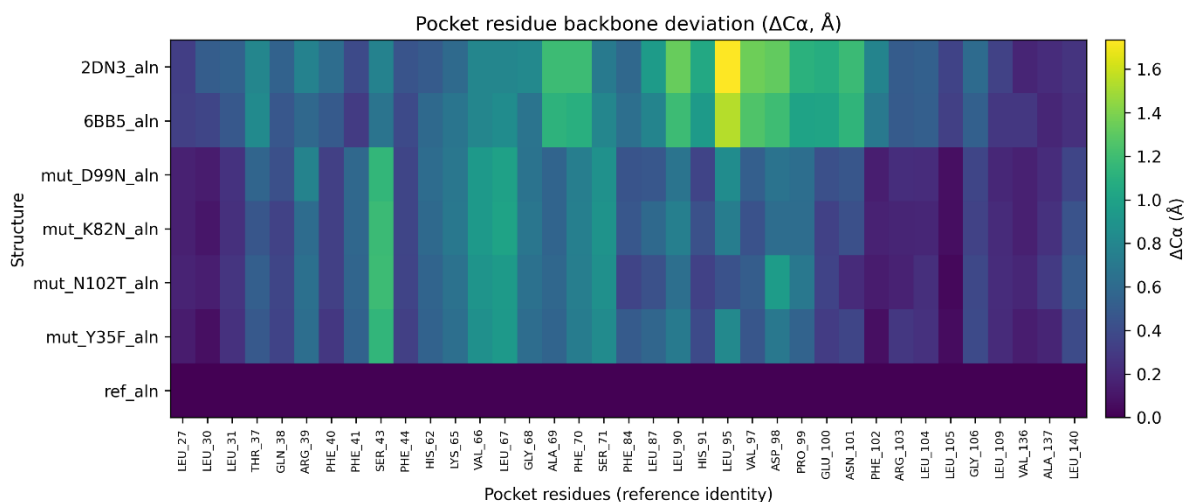


Figure 7: Pocket per-residue backbone deviations across the haemoglobin protein subset

5.4.3. Spatial distribution of residue-level deviations

As seen in Figure 7, residue-level deviations were not uniformly distributed across the heme pocket. Higher deviations were concentrated within specific residue regions, particularly around those corresponding to approximately residues 60-70 and 90-100 in the reference sequence. The other regions of the pocket exhibited consistently lower deviations across all haemoglobin variants.

5.4.4. Most variable pocket residues

The 25 most variable pocket residues were analysed separately in Figure 8. These residues exhibited $\Delta C\alpha$ values within approximately 0.6 to 1.7 Å. Despite being the most variable subset of residues, they remained within a limited deviation range. Furthermore, no residues had large-scale displacement relative to the reference structure.

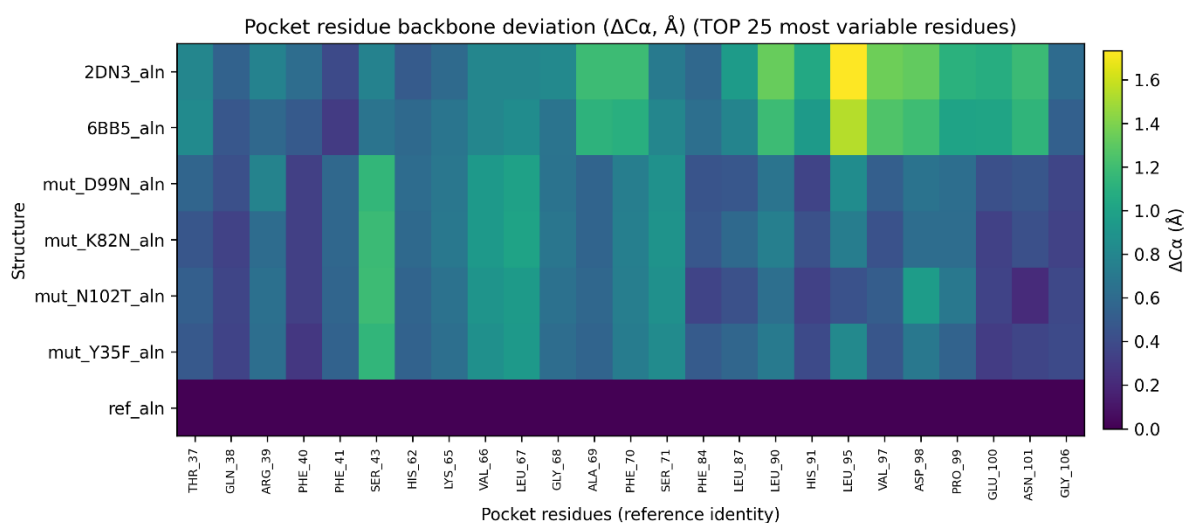


Figure 8: Pocket residue backbone deviations for the 25 most variable pocket residues

6. Discussion

6.1. Conceptual framework

This study aimed to investigate how structural conservation and variation in haemoglobin influence ligand-binding behaviour. In particular, the work attempted to distinguish between structurally conserved and flexible regions of the protein and. Furthermore, it tried to estimate how these structural features relate to ligand-binding energetics, binding mode reproducibility, and spatial constraints of the ligand positioning. To tackle these questions, a blend of blind molecular docking, coordinate-based structural analysis and centroid-based quantification of ligand dispersion was employed.

The analytical framework is based on the postulate that haemoglobin represents a highly conserved protein system, in which functional requirements impose strong structural constraints, particularly in the heme-binding pocket, while allowing a degree of flexibility in regions related to allosteric regulation (Meng et al., 2011). Therefore, incorporating structural metrics (RMSD), energetic descriptors (docking scores), and geometric measures (centroid dispersion), the work provides a multi-level characterisation of protein-ligand interactions.

6.2. Key findings

The results of this study reveal several patterns across the analysed dataset. First, the structural comparison showed a high degree of structural conservation, particularly in the heme-binding pocket region, as both global and local RMSD values remained low. Second, ligand-dependent differences in docking behaviour were observed. The heme-derived ligands consistently exhibited favourable binding energies across all proteins. In contrast, the allosteric ones had greater variability, with 2,3-BPG displaying a broad energy range and IHP consistently yielding weak binding energies. This led to no direct correspondence between binding energy and cluster reproducibility being observed. Systems with strongly favourable binding energy did not necessarily converge to a dominant binding mode, while moderate-energy systems, especially involving 2,3-BPG, often had higher reproducibility. Then, centroid-based analysis further revealed that ligand-dependent spatial constraints dominate over protein-dependent effects. IHP exhibited a highly restricted dispersion, while 2,3-BPG showed a broader, but structured distribution of binding positions. Notably, the differences in centroid dispersion between protein variants were moderate, rather than large-scale. Lastly, across all systems, ligand binding remained confined to known functional regions, reflecting that structural variation affects the extent of positional flexibility rather than the location of binding.

6.3. Ligand-dependent docking behaviour

The analysis of docking behaviour across haemoglobin variants revealed that ligand-specific properties play a dominant role in determining both binding energetics and the reproducibility of binding modes. Structural differences between protein variants led to small variations, but the overall patterns were primarily driven by the chemical nature of the ligands.

Heme-derived ligands consistently exhibited strongly favourable binding energies across all haemoglobin variants. This behaviour is expected, as the heme group represents the native prosthetic ligand of haemoglobin and is accommodated within a highly specialised binding pocket. The geometry and chemical environment of the heme pocket are optimised for stable coordination of the porphyrin ring and its central metal ion (R. Dickerson & Geis, 1993; Perutz

et al., 1960). As a result, the relative uniformity of binding energies across modern, mutant, and ancestral proteins further supports the conclusion that the functional site remains structurally and chemically conserved.

Conversely, the allosteric ligands exhibited more heterogeneous energetic behaviour. 2,3-BPG showed a broad range of binding energies across protein variants, with more favourable values in ancestral proteins and more variable in modern and mutant forms. This variability likely reflects the differences in the electrostatic environment and geometry of the central cavity that moderates 2,3-BPG binding. As this cavity is formed by inter-subunit interactions, even subtle structural differences may influence ligand binding behaviour (Arnone, 1972; Benesch & Benesch, 1967). On the other hand, IHP exhibited consistently weak binding energies across all variants. Given its high negative charge and known role as a strong allosteric effector, this likely reflects limitations of the docking scoring functions in the accurate modelling of highly charged ligands and long-range electrostatic interactions (Meng et al., 2011). Moreover, the size, charge density and flexibility of IHP introduce desolvation and entropic effects that can shift absolute energy values towards less favourable ranges. As a consequence, these values may appear unfavourable compared to other ligands while still representing comparatively favourable binding modes within the context of IHP itself. Furthermore, IHP binding depends on the electrostatic environment of the haemoglobin central cavity formed by the quaternary structure. This could potentially contribute to an underestimation of binding strength in some simplified docking models (Arnone & Perutz, 1974; Gilson & Zhou, 2007; Meng et al., 2011; Shulman et al., 1975).

A key observation from the docking analysis is the absence of a consistent relationship between binding energy and cluster reproducibility. Heme ligands displayed strongly favourable binding energies but often had moderate reproducibility, suggesting the presence of multiple energetically similar binding orientations within the pocket. In contrast, 2,3-BPG frequently exhibited higher reproducibility, despite moderate binding energies, indicative of a more geometrically constrained binding mode within the central cavity (Arnone, 1972; Morris et al., 1998; Perutz, 1983). This decoupling of energetic favourability and binding mode reproducibility demonstrates that these metrics capture distinct aspects of protein-ligand interactions. On one hand, binding energy reflects an estimated thermodynamic stability, whereas reproducibility reflects convergence towards a consistent binding geometry. Therefore, the lack of correlation between these measures solidifies that they have to be treated as separate criteria when interpreting molecular docking results.

In general, these findings propose that ligand identity is the primary determinant of docking behaviour, while protein variation introduces more subtle modulation. Heme ligands are governed by strong structural compatibility with a conserved binding pocket, whereas allosteric ligands are more sensitive to variations in electrostatics and cavity geometry. This reinforces the idea of haemoglobin as a structurally robust system with limited, but functionally relevant flexibility in regulatory regions (Eaton et al., 1999; Hardison, 1996; Perutz, 1970).

6.4. Spatial constraints of ligand binding

The centroid-based analysis provided insight into the spatial constraints governing ligand binding. As opposed to the energetic trends described above, the spatial distribution of ligand positions revealed a strong dependence on ligand type, with limited influence of protein variation.

The IHP ligand exhibited highly restricted spatial dispersion across the whole dataset, with centroid positions confined to a narrow range. This indicates a highly constrained binding mode, consistent with the highly charged nature of IHP (Arnone & Perutz, 1974; Shulman et al., 1975). Additionally, the uniform dispersion across modern, mutant, and ancestral proteins further suggests that the structural features that govern IHP binding are conserved and set strict positional constraints on the ligand. On the contrary, 2,3-BPG displayed a broader range of values, suggesting greater positional variability within the same binding region. The increased dispersion likely reflects weaker electrostatic anchoring and greater sensitivity to subtle differences in cavity geometry and residue positioning. Again, small structural variation can influence ligand positioning without altering the binding region (Benesch & Benesch, 1967; Perutz, 1983).

Nevertheless, ligand binding remained spatially localised within the central cavity. No alternative binding regions were observed, and the centroid clouds consistently occupied the same general location. Therefore, the observed variation reflects differences in the extent of spatial spread rather than shifts in the binding site. Notably, the magnitude of variation in centroid dispersion between proteins was moderate. Even though ancestral and some mutant proteins exhibited greater dispersion for 2,3-BPG than modern ones, these differences remained within a limited range and did not result in distinct or separated binding clusters. This puts forward the idea that structural variation influences the degree of flexibility within the binding region rather than fundamentally altering the ligand-binding behaviour.

Altogether, these findings reinforce the distinction between energetic and geometric aspects of protein-ligand interactions. While docking energies reflect estimated binding favourability, centroid dispersion captures the extent to which a binding site spatially restricts ligand positioning. The strong ligand dependence observed in the centroid analysis further supports the conclusion that haemoglobin maintains a conserved binding architecture, in which ligand-specific properties determine the degree of positional flexibility.

6.5. Structural basis of haemoglobin function

The structural exploration revealed a high degree of conservation, especially within the heme-binding pocket. Both global and local RMSD values remained in a narrow range, with pocket-specific deviations being consistently lower than the global ones. Such a pattern reflects the strong functional constraints imposed on the globin fold, where precise positioning of residues is required for the effective coordination of the prosthetic group and reversible oxygen binding (Hardison, 1996; Perutz et al., 1960). This conservation, therefore, suggests that the fundamental architecture of the binding site is maintained despite sequence variation.

At the residue level, structural variability was not uniformly distributed but rather localised to specific regions of the protein. Increased deviations were observed in parts corresponding to approximately residues 60-70 and 90-100, which likely represent loop regions or peripheral helices. These regions are known to exhibit greater conformational flexibility and are often involved in inter-subunit interactions and allosteric transitions. However, residues directly related to the heme-binding environment displayed consistently low deviations, further solidifying the stability of the functional core (Eaton et al., 1999; Perutz, 1970).

Overall, the findings support a model in which haemoglobin maintains its function through a conserved structural framework complemented by localised flexibility in peripheral regions.

The rigid preservation of the heme-binding pocket ensures reliable ligand coordination, while variability in the surrounding regions might enable subtle adjustments in protein conformation.

6.6. Structure-function relationship

The combination of structural and docking analyses provides a coherent framework of how haemoglobin maintains functional stability while allowing limited modulation of ligand-binding behaviour. First, the structural analysis (Section 6.5) demonstrated that the heme-binding pocket is highly conserved across modern and mutant variants, with low RMSD values indicating minimal deviation in the core functional architecture. As suggested earlier, this conservation reflects the strong evolutionary constraints, where the precise geometry of the heme environment is essential for oxygen binding (Hardison, 1996; Perutz, 1970).

Then, within this conserved structural framework, the docking results of Section 6.3 revealed that the ligand identity is the primary determinant of binding behaviour, while protein variation introduces more subtle modulation. Moreover, the absence of a consistent relationship between binding energy and reproducibility indicates that ligand binding reflects an interplay between thermodynamic and geometric factors. In particular, the presence of multiple energetically similar binding modes for heme ligands contrasts with the more consistent binding behaviour observed for 2,3-BPG. This highlights the differences in how binding sites constrain ligand positioning.

Finally, the centroid dispersion analysis (Section 6.4) further refined this interpretation by demonstrating that ligand binding remains spatially localised within conserved functional regions across all variants. The differences between proteins are expressed as variations in positional flexibility within these regions. The contrast between the highly constrained binding of IHP and the broader dispersion of 2,3-BPG highlights that ligand-specific properties determine how strongly a binding site restricts ligand motion. The moderate magnitude of variation across proteins suggests that structural differences influence flexibility without disrupting the underlying binding architecture.

Altogether, these findings support a model in which haemoglobin function arises from a conserved structural core that constrains ligand binding, combined with limited flexibility that modulates ligand binding. The structural conservation defines the permissible binding space, while ligand-specific and subtle structural variation shape the energetic and geometric characteristics of binding. This interpretation is consistent with the established view of haemoglobin as an allosteric protein in which function arises from small, distributed conformational adjustments (Eaton et al., 1999).

6.7. Biological implications and comparison with literature

The findings of this study are consistent with established structural and functional properties of haemoglobin. The strong conservation of the heme-binding pocket aligns with previous structural studies, demonstrating that functional requirements tightly conserve the globin fold. This supports the idea that haemoglobin function relies on a stable structural core that tolerates only limited perturbation (Hardison, 1996). The ligand-dependent differences observed in docking behaviour are also in agreement with known mechanisms of allosteric regulation. The central cavity serves as a binding site for effectors, where electrostatic interactions modulate oxygen affinity. And the sensitivity of 2,3-BPG binding to structural variation, as reflected in both energetic and spatial analyses, is consistent with its role as a regulator, whose interaction

depends on subtle features of cavity geometry (Arnone, 1972; Benesch & Benesch, 1967; Perutz, 1983). At the same time, the spatial restriction observed for IHP supports findings which show that highly charged polyanions can stabilise haemoglobin through dense electrostatic interactions (Arnone & Perutz, 1974; Shulman et al., 1975). The blend of constrained spatial positioning and weak docking energies highlights the importance of considering electrostatic context and solvent effects when interpreting computational predictions for ligands (Gilson & Zhou, 2007; Meng et al., 2011).

6.8. Limitations and future directions

Several limitations of this study should be considered. First, the molecular docking approach relies on empirical scoring functions that provide only an estimate of binding affinity. Such functions are known to inadequately capture long-range electrostatic interactions and solvent effects, which are critical for highly charged ligands like IHP (Gilson & Zhou, 2007; Meng et al., 2011). Second, the use of blind docking allows unbiased exploration of the protein surface, but it could yield multiple energetically similar binding poses, especially for ligands interacting with large or flexible regions. Third, the structural models here represent static conformations and do not account for dynamic behaviour. Haemoglobin function is inherently linked to conformational transitions between different states, which influence both ligand accessibility and binding affinity. Therefore, the absence of explicit conformational sampling limits the ability to fully capture the range of physiologically relevant interactions (Eaton et al., 1999; Perutz, 1970). Additionally, the structural comparison was restricted to modern and mutant proteins. Even though this still ensured robust comparative analysis, it constrained the ability to conclude the structural variability across evolutionary timescales.

It is encouraged that future works address these limitations by incorporating molecular dynamics simulations to capture conformational flexibility and by applying more advanced free energy methods to improve binding affinity estimation. Furthermore, expanding the dataset to include additional variants would strengthen comparative analysis. Lastly, experimental validation of predicted binding behaviour would provide an important basis for assessing the reliability of computational approaches.

7. Conclusion

This study aimed to investigate how structural conservation and variation in haemoglobin influence ligand-binding behaviour. This was achieved by integrating structural analysis, molecular docking, and centroid-based spatial evaluation. The results demonstrated that the heme-binding pocket remains highly conserved, providing a stable structural framework for ligand coordination. Additionally, it was observed how ligand-specific properties govern binding energetics and spatial behaviour. In particular, the separation of binding energy and reproducibility, together with ligand-dependent differences in centroid dispersion, strongly suggest that protein-ligand interactions reflect a combination of thermodynamic and geometric constraints. These findings render the study's aims achieved, as they provided a consistent structure-function interpretation in which haemoglobin maintains functional robustness through conservation of its core architecture, while still allowing limited modulation of ligand flexibility. Consequently, this has broader implications for understanding allosteric regulation and for the interpretation of docking results in complex protein systems.

8. References

- Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Ballard, A. J., Bambrick, J., Bodenstein, S. W., Evans, D. A., Hung, C. C., O'Neill, M., Reiman, D., Tunyasuvunakool, K., Wu, Z., Žemgulytė, A., Arvaniti, E., ... Jumper, J. M. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* 2024 630:8016, 630(8016), 493–500. <https://doi.org/10.1038/s41586-024-07487-w>
- Arnone, A. (1972). X-ray Diffraction Study of Binding of 2,3-Diphosphoglycerate to Human Deoxyhaemoglobin. *Nature* 1972 237:5351, 237(5351), 146–149. <https://doi.org/10.1038/237146a0>
- Arnone, A., & Perutz, M. F. (1974). Structure of inositol hexaphosphate–human deoxyhaemoglobin complex. *Nature* 1974 249:5452, 249(5452), 34–36. <https://doi.org/10.1038/249034a0>
- Benesch, R., & Benesch, R. E. (1967). The effect of organic phosphates from the human erythrocyte on the allosteric properties of hemoglobin. *Biochemical and Biophysical Research Communications*, 26(2), 162–167. [https://doi.org/10.1016/0006-291X\(67\)90228-8](https://doi.org/10.1016/0006-291X(67)90228-8)
- Bunn, H. Franklin., & Forget, B. G. . (1986). *Hemoglobin--molecular, genetic, and clinical aspects*. 690.
- Dickerson, R. E., & Geis, I. (1993). Hemoglobin: Structure, function, evolution, and pathology. *The Benjamin/Cummings Publishing Company, Menlo Park, CA*. <https://www.scirp.org/reference/referencespapers?referenceid=405479>
- Dickerson, R., & Geis, I. (1993). Hemoglobin structure, function, evolution and pathology. *Benjamin-Cummings*.
- Eaton, W. A., Henry, E. R., Hofrichter, J., & Mozzarelli, A. (1999). Is cooperative oxygen binding by hemoglobin really understood? *Nature Structural Biology*, 6(4), 351–358. <https://doi.org/10.1038/7586>
- Gilson, M. K., & Zhou, H. X. (2007). Calculation of protein-ligand binding affinities. *Annual Review of Biophysics and Biomolecular Structure*, 36, 21–42. <https://doi.org/10.1146/annurev.biophys.36.040306.132550>
- Gowers, R. J., Linke, M., Barnoud, J., Reddy, T. J. E., Melo, M. N., Seyler, S. L., Domański, J., Dotson, D. L., Buchoux, S., Kenney, I. M., & Beckstein, O. (2016). MDAnalysis: A Python Package for the Rapid Analysis of Molecular Dynamics Simulations. *SciPy 2016*, 98–105. <https://doi.org/10.25080/majora-629e541a-00e>
- Hanwell, M. D., Curtis, D. E., Lonie, D. C., Vandermeersch, T., Zurek, E., & Hutchison, G. R. (2012). Avogadro: an advanced semantic chemical editor, visualization, and analysis platform. *Journal of Cheminformatics* 2012 4:1, 4(1), 17-. <https://doi.org/10.1186/1758-2946-4-17>

- Hardison, R. C. (1996). A brief history of hemoglobins: Plant, animal, protist, and bacteria. *Proceedings of the National Academy of Sciences of the United States of America*, 93(12), 5675–5679. <https://doi.org/10.1073/pnas.93.12.5675>
- Hochberg, G. K. A., & Thornton, J. W. (2017). Reconstructing Ancient Proteins to Understand the Causes of Structure and Function. *Annual Review of Biophysics*, 46, 247–269. <https://doi.org/10.1146/annurev-biophys-070816-033631>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science and Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- Kendrew, J. C., Dickerson, R. E., Strandberg, B. E., Hart, R. G., Davies, D. R., Phillips, D. C., & Shore, V. C. (1960). Structure of Myoglobin: A Three-Dimensional Fourier Synthesis at 2 Å. Resolution. *Nature* 1960 185:4711, 185(4711), 422–427. <https://doi.org/10.1038/185422a0>
- Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B. A., Thiessen, P. A., Yu, B., Zaslavsky, L., Zhang, J., & Bolton, E. E. (2025). PubChem 2025 update. *Nucleic Acids Research*, 53(D1), D1516–D1525. <https://doi.org/10.1093/nar/gkae1059>
- Meng, X.-Y., Zhang, H.-X., Mezei, M., & Cui, M. (2011). Molecular docking: a powerful approach for structure-based drug discovery. *Current Computer-Aided Drug Design*, 7(2), 146–157. <https://doi.org/10.2174/157340911795677602>
- Morris, G. M., Goodsell, D. S., Halliday, R. S., Huey, R., Hart, W. E., Belew, R. K., & Olson, A. J. (1998). Automated Docking Using a Lamarckian Genetic Algorithm and an Empirical Binding Free Energy Function. *Journal of Computational Chemistry*, 19(14).
- National Center for Biotechnology Information (NCBI)[Internet]. (1988). Bethesda (MD): National Library of Medicine (US), National Center for Biotechnology Information; National Library of Medicine. <https://www.ncbi.nlm.nih.gov/>
- Olson, J. S., & Phillips, G. N. (1997). Myoglobin discriminates between O₂, NO, and CO by electrostatic interactions with the bound ligand. *Journal of Biological Inorganic Chemistry*, 2(4), 544–552. <https://doi.org/10.1007/s007750050169>
- Perutz, M. F. (1970). Stereochemistry of cooperative effects in haemoglobin. *Nature*, 228(5273), 726–734. <https://doi.org/10.1038/228726a0>
- Perutz, M. F. (1983). Species adaptation in a protein molecule. *Molecular Biology and Evolution*, 1(1), 1–28. <https://doi.org/10.1093/oxfordjournals.molbev.a040299>
- Perutz, M. F., Rossmann, M. G., Cullis, A. F., Muirhead, H., Will, G., & North, A. C. T. (1960). Structure of Hæmoglobin: A Three-Dimensional Fourier Synthesis at 5.5-Å. Resolution, Obtained by X-Ray Analysis. *Nature* 1960 185:4711, 185(4711), 416–422. <https://doi.org/10.1038/185416a0>
- Pettersen, E. F., Goddard, T. D., Huang, C. C., Meng, E. C., Couch, G. S., Croll, T. I., Morris, J. H., & Ferrin, T. E. (2020). UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Science : A Publication of the Protein Society*, 30(1), 70. <https://doi.org/10.1002/pro.3943>

- Pillai, A. S., Chandler, S. A., Liu, Y., Signore, A. V., Cortez-Romero, C. R., Benesch, J. L. P., Laganowsky, A., Storz, J. F., Hochberg, G. K. A., & Thornton, J. W. (2020). Origin of complexity in haemoglobin evolution. *Nature*, 581(7809), 480–485. <https://doi.org/10.1038/s41586-020-2292-y>
- Posit team (2025). (n.d.). *RStudio: Integrated Development Environment for R*. Posit Software, PBC. Retrieved March 29, 2026, from <http://www.posit.co/>
- Sanner, M. (1999). Python: a programming language for software integration and development. *Journal of Molecular Graphics and Modelling*.
- Shulman, R. G., Hopfield, J. J., & Ogawa, S. (1975). Allosteric interpretation of haemoglobin properties. *Quarterly Reviews of Biophysics*, 8(3), 325–420. <https://doi.org/10.1017/S0033583500001840>
- Thom, C. S., Dickson, C. F., Gell, D. A., & Weiss, M. J. (2013). Hemoglobin variants: biochemical properties and clinical correlates. *Cold Spring Harbor Perspectives in Medicine*, 3(3). <https://doi.org/10.1101/cshperspect.a011858>
- Thornton, J. W. (2004). Resurrecting ancient genes: Experimental analysis of extinct molecules. *Nature Reviews Genetics*, 5(5), 366–375. <https://doi.org/10.1038/nrg1324>
- Warren, G. L., Andrews, C. W., Capelli, A. M., Clarke, B., LaLonde, J., Lambert, M. H., Lindvall, M., Nevins, N., Semus, S. F., Senger, S., Tedesco, G., Wall, I. D., Woolven, J. M., Peishoff, C. E., & Head, M. S. (2005). A Critical Assessment of Docking Programs and Scoring Functions. *Journal of Medicinal Chemistry*, 49(20), 5912–5931. <https://doi.org/10.1021/jm050362n>